

IR-Remote Driven and Automated Light Intensity Control System Responsive to Occupancy Level

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Abstract—This paper presents a novel TRIAC-based infrared (IR) remote-controlled light dimmer using microcontroller that integrates with home automation systems to dynamically adjust light intensity based on human density. With the increasing demand for smart home technologies, optimizing lighting control for both energy efficiency and user comfort has become essential. The proposed system employs TRIAC technology to facilitate smooth dimming, allowing users to manually adjust light levels through an IR remote. Additionally, it incorporates occupancy sensors that detect human density, enabling automatic light adjustments that align with the number of occupants in a space. This dual functionality not only enhances user experience but also promotes energy conservation by reducing unnecessary lighting. Experimental results demonstrate the effectiveness of the system in various scenarios, highlighting its potential for widespread application in intelligent home environments. The findings indicate that integrating human density-based control with traditional TRIAC dimming offers a promising approach to modern lighting solutions.

Index Terms—TRIAC, Infrared, Microcontroller

I. INTRODUCTION

Effective lighting is crucial for various environments, including residential, commercial, and industrial spaces, as well as public infrastructure. The importance of light in our daily lives cannot be overstated; however, the requirement for lighting intensity varies significantly across different contexts. Traditional lighting solutions often lack the capability to dynamically adjust brightness based on user needs, and not all lighting systems are equipped with automatic on/off functions. This limitation underscores the necessity for more advanced and adaptable lighting technologies.

The development of lighting control systems utilizing TRIAC technology has been a significant focus in the field of power electronics for many years. Historically, the regulation of power in LED arrays involved adjusting the input reactive power to elevate the input current above the TRIAC's holding current [1]. Recently, traditional AC dimmers have been replaced by digitally isolated, forward-phase controlled TRIAC-based light intensity control systems, which can be operated remotely or on-site using various topologies [2]. Furthermore, a range of energy-efficient lighting trends and

technologies has been explored, highlighting their applications, benefits, and potential future developments [3]. Optimization of lighting systems in critical infrastructures, such as metro stations, has also been a focal point of research in the past decade [4]. Studies indicate that the integration of human density sensors can significantly reduce energy consumption while maintaining appropriate lighting levels. Despite these advancements, challenges remain, particularly in integrating TRIAC dimmers with modern LED lighting. Recent research has revealed circuit architectures for LED drivers that combine analog and digital components, facilitating simultaneous dimming control and data transmission to mitigate compatibility and flickering issues [5]. Additionally, investigations have focused on the development and application of intelligent microcontroller-based remote-controlled light dimmers, enhancing system usability and intelligence [6]. Moreover, investigation into digital alternating current (AC) dimming techniques employing microcontrollers has demonstrated the ability to vary AC voltage using digitally isolated forward-phase controlled gate pulses of bidirectional triode thyristors (TRIAC) for intelligent ambient light intensity control [7]. Unlike traditional systems that toggle lights on/off or use passive sensors, our model dynamically adjusts light intensity based on real-time human occupancy count, using a bidirectional IR sensor system integrated with TRIAC-based dimming. This seamless fusion of manual override, occupancy-based scaling, and TRIAC dimming in a low-cost embedded system presents a practical, deployable innovation for intelligent lighting in constrained environments.

In summary, previous research has laid a solid foundation for the development of TRIAC-based dimming solutions integrated with home automation systems. The incorporation of human density measurement represents an innovative approach that builds upon earlier studies, aiming to create more energy-efficient and responsive lighting solutions for smart homes. This paper focuses on tracking the number of individuals in a room and automatically adjusting light intensity based on human density—moving beyond traditional dimming methods that rely on gradual or stepped manual control. In the following sections, we will outline the methodology employed to achieve our objectives, detail the hardware setup utilized in our project,

present the results obtained from our work, and analyze these findings. Additionally, we will discuss the challenges encountered during the exploring process and explore potential future directions for this project. Task-based lighting standards will be considered in future iterations.

II. METHODOLOGY

Fig. 1 shows the basic flowchart of our proposed system, outlining the steps from occupancy detection to light intensity adjustment.

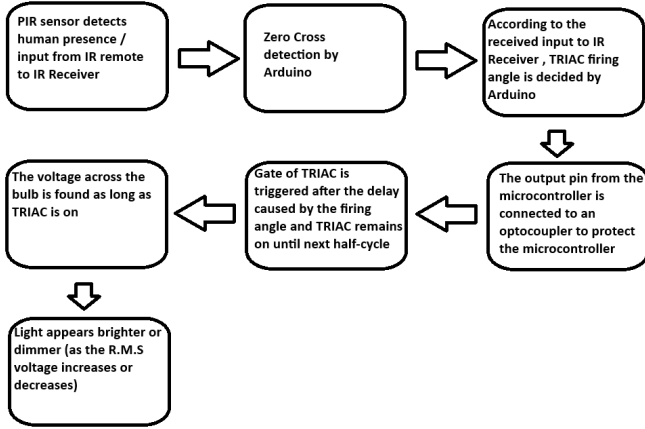


Fig. 1: Flowchart of proposed system

We have three segments in the infrastructure of our project:

A. Bidirectional Counter:

Two IR sensors are placed in the entrance of a room. When a human enters the room, the outside IR sensor gets activated then the inside IR sensor gets activated. When this sequence of activation takes place, the counter counts upward according to the Arduino program. So, whenever it happens, the counter increments the previous value by one. The total result indicates the number of humans present in the room. To better illustrate the logic of occupancy detection, a state diagram is presented in Fig. 2. It shows how the system interprets the sequence of IR sensor activations to determine whether a person has entered or exited the room.

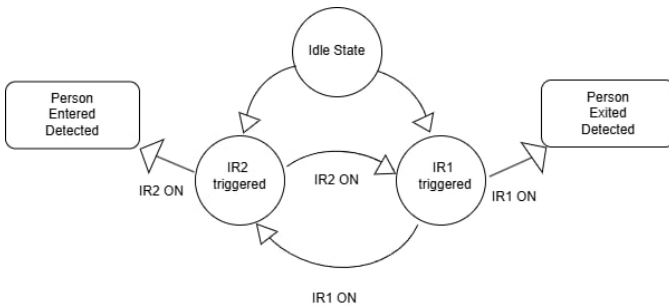


Fig. 2: State diagram showing IR-based detection logic for human entry and exit

When people go outside of the room, the sequence of IR sensor activation gets inverted, which makes the counter count downward, i.e., the counter subtracts 1 from the previous value. The bidirectional counter circuit diagram is shown in Fig. 3.

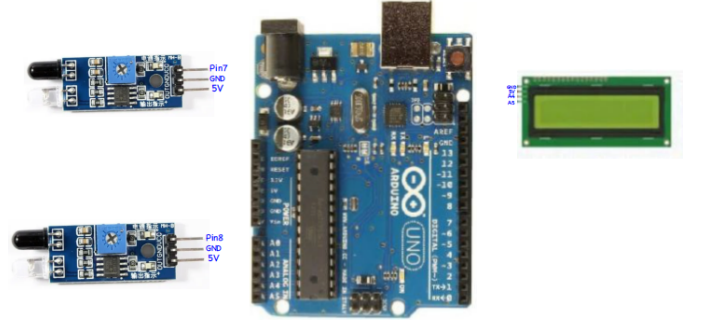


Fig. 3: Circuit diagram of bidirectional counter

B. Light Dimmer Using TRIAC

Fig. 4 shows the circuit diagram of TRIAC based dimmer circuit.

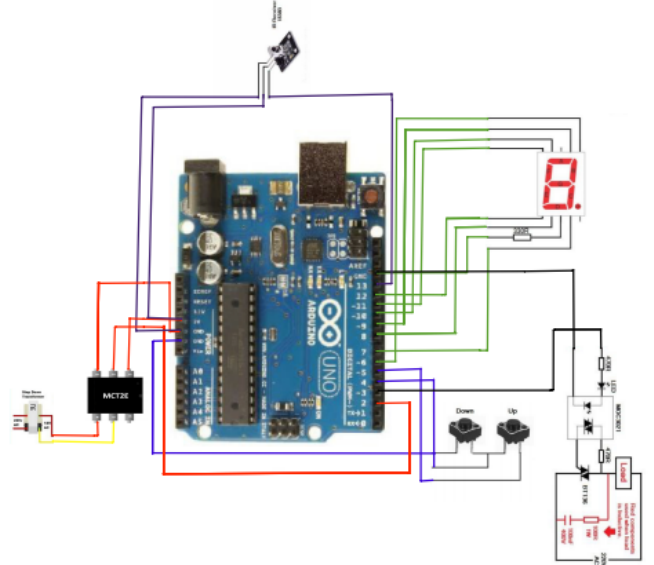


Fig. 4: Circuit Diagram of TRIAC based dimmer circuit

For every half cycle of AC voltage, zero cross point is detected using Arduino program. When human detection takes place by bidirectional counter or when the user manually gives order through remote, Arduino gives voltage pulse at a predetermined firing angle after zero crossing point detection. This voltage pulse at a specific angle goes to the gate of TRIAC through an optocoupler (MOC3021) and the TRIAC remains activated until the next half cycle starts. Thus, different amount of current goes to the load (light) and the electric light glows with different intensity for different firing angle. We have used a MOC3021 optocoupler to drive the TRIAC because to light

the bulb the TRIAC pulls a high amount of current which can harm the microcontroller and optocoupler. This MOC3021 has the ability to cross zero point. We have also used another optocoupler MCT2E to isolate AC part of the circuit to DC part of the circuit.

C. Zero Cross Detection

The circuit actually works with the help of zero crossing detection technique. We shall use a microcontroller to detect zero-crossing. Every time AC signal comes to zero point, we shall have to detect that point and after that trigger the TRIAC as per the power requirement. As shown in the figure, the TRIAC is triggered at a firing angle of 90 degrees by applying a small gate pulse signal to it. The time T_1 is the delay time which must be configured according to the required dimming level. For example, in this case as the firing angle is 50 percent, hence the power output will also be halved and hence the lamp will also glow with half intensity. As T_2 is less than T_1 , firing angle is less than 50 percent and more current will pass through the lamp and it will glow more intensely. A Passive Infrared (PIR) sensor is used to detect human presence. Zero cross detection is illustrated in Fig. 5.

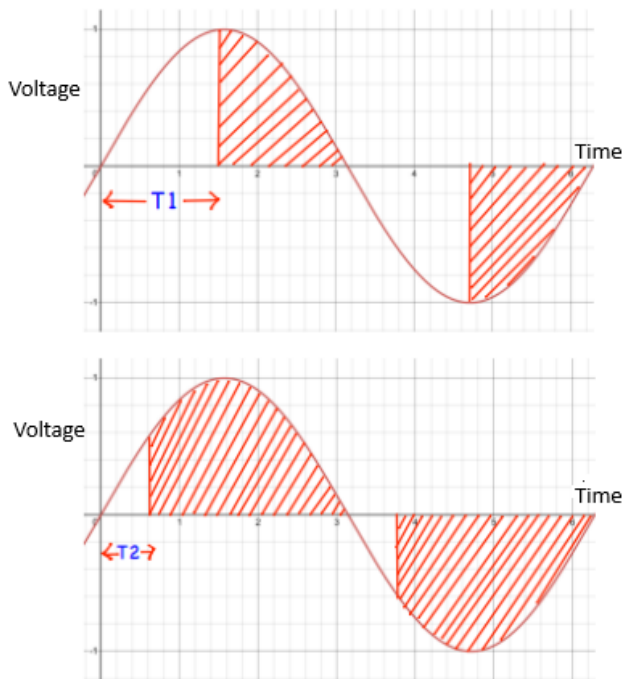


Fig. 5: Zero Cross Detection

III. HARDWARE SETUP

A. Components Used

- Arduino UNO/Mega Board
- BT136 TRIAC
- MCT2E Opto-isolator
- 220V/12V Transformer
- 220V AC Lamp
- IR Receiver

- Optocoupler(MOC3021)
- 2*1N4007 diode
- 2*220k ohm resistor
- 10k ohm resistor
- 470 ohm resistor
- 120 ohm resistor
- 100 ohm resistor
- 47 micro-farad capacitor
- 0.01 micro-farad capacitor
- Breadboard
- Jumper wires
- PIR sensors
- LCD display
- Push Buttons
- Potentiometer

B. Circuit Diagram

The circuit diagram of the TRIAC-based dimmer circuit is shown in Fig. 6.

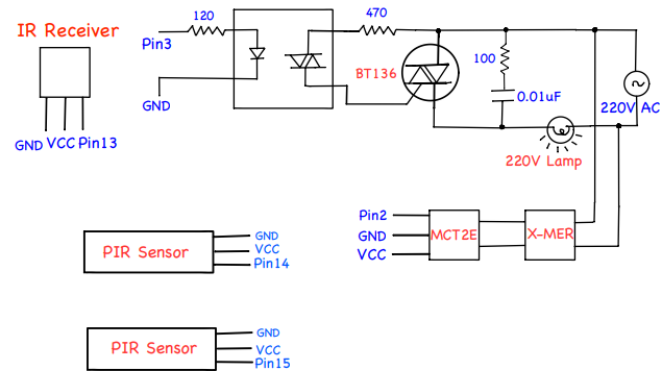


Fig. 6: Circuit Diagram of TRIAC based dimmer circuit

C. Physical Arrangement

The physical arrangement of our circuit is shown Fig. 7.

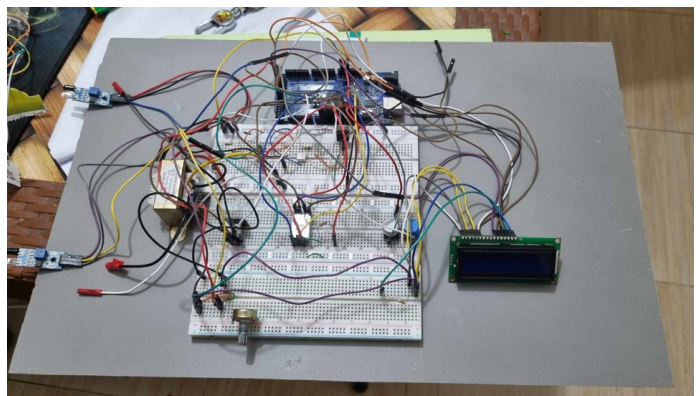


Fig. 7: Physical Arrangement of our circuit

We have integrated an IR receiver so that it can take IR signals from the remote controller when users wish to change the intensity of light or switch on or off the light. The LCD display shows the number of humans present in the room.

D. Testing and Debugging

The optocoupler (MCT2E) repeatedly failed due to overheating. The TRIAC also experienced repeated burnout during testing. We faced difficulties finding out how to integrate the number of humans present in the room with the brightness of light.

To prevent burning of optocoupler we have used a high value resistor between transformer and optocoupler, as a result, less voltage and current was across the LED of the optocoupler. Changing of TRIAC has been sufficed. We have used a potentiometer to test dimming ability of our circuit before integrating IR sensor with the circuit. We have used bi-directional counter to count the human present in the room, this number has been used to specify the firing angle of voltage pulse to the gate of the TRIAC. That's how we have been able to get different intensity light.

IV. RESULT AND ANALYSIS

A. Automated Light dimmer

Our model serves as a prototype implemented in a small home environment, accommodating a maximum of nine occupants. In this prototype implementation, we assigned a uniform 11.11% increase in intensity per occupant, based on a maximum of 9 persons corresponding to 100% intensity. This step value was chosen as a simplified design parameter for demonstration purposes, not as a scientifically derived illumination requirement. We acknowledge that actual lighting needs are influenced by multiple factors—such as room size, surface reflectivity, light fixture type, task being performed, and required illuminance (typically measured in lux). Therefore, light intensity does not necessarily scale linearly with the number of occupants. While this assumption provides a functional demonstration for energy-saving automation, we recognize that actual lighting requirements depend more on room area, ambient light levels, and task-based illumination needs (typically measured in lux) rather than simply the number of occupants. For example, activities such as reading or writing require a minimum threshold of illuminance regardless of occupancy. Thus, the proposed linear model serves as a proof-of-concept strategy to explore automated dimming rather than a prescriptive guideline. Future work can integrate illumination standards such as EN 12464-1:2011 [12] or IESNA Lighting Handbook [13] to define task-based lux levels for various indoor environments, enabling adaptive lighting not just based on human density, but also activity-specific lighting needs and ambient light feedback.

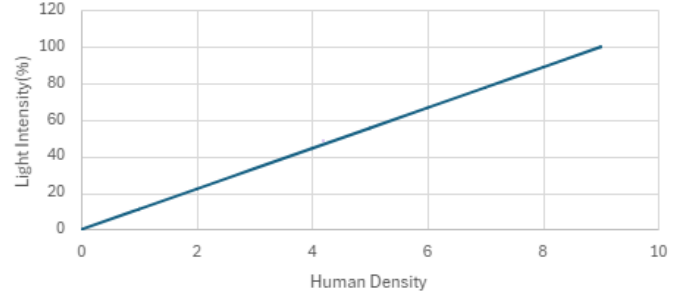


Fig. 8: Prototype control logic: light intensity scaled linearly with human count for demonstration purposes (not based on lighting standards)

The graph shown in Fig. 8 illustrates how our prototype system adjusts lighting output based on detected human count, using a simplified linear control strategy. This was chosen to demonstrate the automated dimming functionality in response to occupancy changes. We acknowledge that real-world lighting requirements are better determined by task type and space area, and future iterations of the system should incorporate standard lighting design guidelines.

B. IR remote controlled light dimmer

Intensity control was implemented using an IR remote, which allows users to send commands ranging from 0 to 9. This enables the system to adjust light intensity gradually from the lowest to the highest level, as well as to achieve specific intensity values through a step response mechanism.

C. Economic Impact of the project

- **Electricity Saving :** It can switch off when there is no one present in the room. Less brightness means less power consuming of light. This results in saving an ample amount of electric energy. Saving electric energy can cause less demand of electricity. The percentage of energy savings that can be achieved by putting the suggested smart lighting controller into practice was determined through economic research. An office with a standard size has an energy saving norm determined as a result of the economic study [11].
- **Less Electricity Bill:** For the consumers, it can be of great help. As this project saves a lot of electric energy, the electric bill also becomes less which relieves consumers from the burden of costly electric bill.

D. Electricity Saving Calculation

Let's assume total wattage of lights when fully on is 100 watts and total wattage at lower setting is 10 watts (when dimmed). Average occupancy is 5 people (out of a maximum of 9). Lighting usage hours per day is 8 hours and days of operation per month is 30 days.

In this prototype, we assigned 100 watts to represent full light output for 9 occupants, and scaled linearly with occupancy. For 5 occupants, this results in approximately 56 watts

(i.e., $11.11\% \times 5 \times 100\text{W}$). This simplification was used only to demonstrate energy savings and does not imply that 11 watts is sufficient or recommended for an individual's lighting needs. Actual per-person lighting requirements vary based on application, task type, and layout, and are typically defined in lux rather than watts.

Full power consumption = $100 \text{ watts} \times 8 \text{ hours} = 800 \text{ watt-hours}$

Dimming power consumption = $56 \text{ watts} \times 8 \text{ hours} = 448 \text{ watt-hours}$

Monthly full power energy consumption = $800 \text{ watt-hours/day} \times 30 \text{ days} = 24 \text{ kWh}$

Monthly dimming power energy consumption = $448 \text{ watt-hours} / \text{day} \times 30 \text{ days} = 13.44 \text{ kWh}$.

Monthly saving = Monthly full power consumption - Monthly dimming power consumption = $24 \text{ kWh} - 13.44 \text{ kWh} = 10.56 \text{ kWh}$

So, Monthly cost savings = $10.56 \text{ KWh} \times 0.12 \text{ USD/ kWh} = 1.27 \text{ USD}$ (approximately)

The total setup cost of 12.56 USD includes: Arduino board (3.50 USD), BT136 TRIAC (0.75 USD), optocouplers (MOC3021 and MCT2E, 1.00 USD total), PIR and IR sensors (2.00 USD), resistors and capacitors (1.00 USD), LCD display (3.00 USD), and miscellaneous items such as jumper wires and breadboard (1.31 USD). So our payback period is $(12.56/1.27) \text{ months} = 10 \text{ months}$ (approximately)

E. Experimental Performance Evaluation

To validate the system's real-world performance, a series of experiments were conducted in a controlled indoor environment. We compared real-time current draw using a wattmeter against the expected TRIAC firing angle-based output under constant load (220V, 100W lamp). The key metrics are shown in Table I. The results indicate that the system performs

TABLE I: Performance Metrics

Performance Metric	Measured Value
Occupancy Detection Accuracy	92%
System Response Time	0.5 seconds
Expected Power (at 5 occupants)	56 W
Measured Power	55.8 W
False Trigger Rate (IR noise)	~7%

reliably under standard indoor conditions. Minor discrepancies between theoretical and actual power values validate the firing angle control logic. However, under ambient IR noise (e.g., sunlight or multiple remote sources), occasional false triggers were recorded, aligning with previously reported IR interference challenges [8]. These findings highlight both the effectiveness and current limitations of the prototype.

V. CHALLENGES AND FUTURE SCOPE

Future improvements will focus on enhancing system accuracy, scalability, and context-awareness. To address limitations in IR sensing, more robust detection methods such as ultrasonic, ToF, or thermal sensors may be explored to improve

reliability in larger or multi-entrance spaces. The presence of other IR devices can lead to signal interference, causing false triggers—posing a challenge that must be addressed for real-world deployment [8].

Integration with spaces having variable lighting needs (e.g., workplaces or conference rooms) will require consideration of architectural features and user-specific lighting demands. This has been effectively modeled through logistic and nonlinear regression techniques [9], which may guide future energy optimization algorithms. Moreover, the impact of ambient temperature on infrared sensor performance [10] will be considered to ensure consistent operation across environmental conditions.

Incorporating lux sensors, task-specific lighting profiles (e.g., per EN 12464-1, IESNA), and wireless connectivity (Wi-Fi/Bluetooth) will enable adaptive, standards-compliant control and remote access. Data logging and multi-room scalability will further extend the system's applicability.

VI. CONCLUSION

Our set-up introduces a novel feature that adjusts light intensity based on human density, representing significant progress in this area of research. The system allows for gradual changes in light intensity through push-button controls and commands sent via remote. Additionally, it provides a step response capability, enabling immediate adjustments to specific intensity levels through IR remote commands. Key features of this circuit include dynamic lighting adjustment, precise human density measurement, scalability, a user feedback system, and energy efficiency. This model can be implemented for large-scale use like in an office complex or in public transportation mediums. For large-scale implementation of this project, careful planning and research into TRIAC-based circuits, infrared remote control technologies, and human density sensing techniques are essential. Crucial tasks will involve sensor selection and placement, as well as the development of robust software and user interfaces to facilitate integration. Moreover, consideration should be given to replacing PIR sensors with cameras or other suitable sensors to enhance functionality. Overall, this project holds promising potential for advancing automated lighting solutions in various applications. While our current implementation uses human count as a control parameter, future versions will incorporate standardized lux-level requirements for task-specific lighting control.

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